

Review Article

Executive functioning in individuals with generalized anxiety disorder: A systematic review and meta-analysis

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ABSTRACT

While deficits in ‘cold’ executive functioning (EF) have been linked to Generalized Anxiety Disorder (GAD), no study has systematically examined whether specific EF domains are disproportionately impaired. Addressing this gap is crucial for clarifying the role of EF in the persistence of anxiety and cognitive inefficiencies associated with GAD. This systematic review and meta-analytic investigation synthesized data from 13,084 participants across 32 studies to provide a comprehensive neuropsychological profile of GAD across the lifespan. Studies exploring GAD diagnosis and GAD symptoms/severity revealed similar patterns, such that, GAD was associated with poorer cognitive flexibility and working memory, but not inhibitory control. Effect sizes were similar across youths and young/middle-aged adults; however, studies involving older adults yielded no association between GAD and any EF domain. Furthermore, effect sizes were notably stronger for clinical samples compared to non-clinical samples, and for self-report measures of EF compared to accuracy- and reaction time-based indices. Further exploration revealed that GAD was linked to slower reaction times on working memory tasks and poorer accuracy on both cognitive flexibility and working memory tasks, challenging attentional control theory which suggests that anxiety impacts cognitive performance efficiency (reaction time)—not accuracy. An updated framework specific to GAD is therefore needed to clarify the underlying cognitive mechanisms underlying this condition. These findings have critical implications for treatment, as targeted interventions designed to strengthen impaired EF domains could enhance anxiety management and improve overall cognitive functioning and well-being for individuals with GAD.

1. Introduction

Anxiety disorders, including generalized anxiety disorder (GAD), are among the most prevalent and comorbid neuropsychiatric conditions across all age groups (Baldaçara et al., 2024; Portman, 2009). Epidemiological studies indicate that the lifetime prevalence of GAD surpasses that of all other anxiety disorders (Kessler et al., 2001; Mennin et al., 2008; Noyes Jr, 2001; Somers et al., 2006). Additionally, GAD is associated with a high burden of disability, with 56 % of individuals diagnosed with the disorder experiencing severe impairment—exceeding the disability rates of other anxiety disorders (Kessler et al., 2009). The impact of GAD is widespread and prevalent across the lifespan, being one of the most common psychiatric conditions among children and adolescents (Imran et al., 2017; Kim et al., 2019), young adults (Leonard and Abramovitch, 2019; Rabner et al., 2024), and older adults

(Beaudreau et al., 2013; Luo et al., 2022).

According to the DSM-5, GAD is characterized by excessive, persistent worry about various aspects of life (e.g., work, school, responsibilities) that is difficult to control and accompanied by at least one physical symptom for six months or longer (American Psychiatric Association, 2013). As a chronic and often debilitating disorder, GAD is associated with significant physical complaints and cognitive impairments (Zainal and Newman, 2018). Pathological worry in GAD has been linked to deficits in cognitive functioning, which can negatively affect learning, occupational performance, and interpersonal relationships (Zainal and Newman, 2018). Moreover, GAD has been implicated in increased risk for lower life expectancy, cardiovascular disease, and suicidal behavior (Baldaçara et al., 2024; Lenze and Wetherell, 2011; Trindade et al., 2021), as well as impairments in social, occupational, and family functioning (Essau et al., 2014; Przeworski et al., 2011; Swan

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and Kendall, 2016).

Despite the growing body of research on anxiety disorders, the nature and extent of neurocognitive deficits associated with GAD—particularly in executive functioning—remain unclear due to inconsistent findings across studies (Imran et al., 2017; Leonard and Abramovitch, 2019; Luo et al., 2022; Majeed et al., 2023; Portman, 2009; Rabner et al., 2024). Further investigation is needed to clarify the neurocognitive profile of GAD, which may help refine diagnostic approaches and inform targeted interventions.

1.1. Impact of GAD on cognitive and executive functioning

GAD has been consistently associated with impairments in ‘cold’ (non-affective) executive functioning. Executive functions (EFs) are higher-order cognitive processes essential for goal-directed behavior, with three core EFs prominently being identified within the literature: cognitive flexibility, inhibitory control, and working memory (Diamond, 2013; Miyake et al., 2000). These functions support the regulation of thoughts, emotions, and actions, and are considered critical for psychological well-being and daily functioning (Zainal and Newman, 2022). Given that EFs form the foundation for other cognitive processes (Diamond, 2013), disruptions in EF are likely to cascade into broader impairments across multiple cognitive domains (e.g., attention, memory) which have been observed across the literature (e.g., Asfour et al., 2024; Moon and Jeong, 2017).

Theoretical frameworks such as *attentional control theory* propose that high levels of anxiety impair executive functioning by biasing attention toward perceived threats, thereby reducing the availability of cognitive resources for goal-directed processing and diminishing cognitive control and flexibility (Eysenck et al., 2007). According to this theory, anxiety disrupts the balance between the goal-directed (top-down) and stimulus-driven (bottom-up) attentional systems, leading to difficulty shifting attention, suppressing intrusive thoughts, or holding information in mind under emotional stress (Leonard and Abramovitch, 2019; Majeed et al., 2023). Importantly, attentional control theory predicts that anxiety is more likely to impair the *efficiency* of performance (e.g., slower reaction times, greater mental effort) rather than accuracy, as anxious individuals may compensate by investing additional cognitive effort to maintain performance levels. Neurobiological models further suggest that dysregulation in the amygdala and prefrontal cortex underlies both emotional reactivity and EF disruption in GAD (Asfour et al., 2024; Liu et al., 2015).

Despite clear theoretical and neurobiological links, empirical findings on EF in GAD remain inconsistent. Some studies report broad EF impairments, while others show intact functioning or selective deficits limited to specific domains (Baumel et al., 2022; Gualtieri and Morgan, 2008; Gulpers et al., 2022; Tang et al., 2022; Tempesta et al., 2013). A recent systematic review of cognitive function in youth with GAD further highlights this variability, finding deficits in non-verbal memory, receptive language, and motor skills but intact attention, visuospatial ability, and inhibitory control (Rabner et al., 2024). In contrast, other studies have reported poorer attention and inhibitory control in adult samples (e.g., Asfour et al., 2024). These findings reinforce the idea that cognitive performance in GAD may be domain-specific and age-dependent. Moreover, some evidence aligns with attentional control theory, demonstrating that individuals with GAD maintain task accuracy but perform more slowly, reflecting reduced processing efficiency rather than global impairment (Majeed et al., 2023). However, some studies have reported reduced accuracy, contradicting the central tenet of attentional control theory (e.g., Luo et al., 2022; Unterrainer et al., 2018).

Given that many GAD interventions aim to target maladaptive cognitive processes (Holmes et al., 2014), a more refined understanding of how GAD affects EFs is essential. A systematic, domain-specific synthesis of existing findings is therefore warranted to clarify whether and how different EF components are impaired in GAD, and to inform the

development of targeted interventions grounded in up-to-date empirical evidence.

1.2. Syntheses of cognitive functioning in GAD

Gkintoni and Ortiz (2023) conducted a systematic review examining the neuropsychological profile of GAD, identifying potential deficits in attention and EFs, along with alterations in neural structures such as the amygdala and insula. They also noted that comorbid conditions, such as major depressive disorder, may contribute to neuropsychological dysfunction. However, while their review provides valuable insights, the lack of a quantitative synthesis limits its conclusions. Without a meta-analytic approach, it remains unclear how consistent these deficits are across studies or whether they vary by cognitive domain, symptom severity, or sample characteristics.

A recent meta-analysis by Majeed et al. (2023) examined EFs across all anxiety disorders, finding that individuals with anxiety disorders exhibited slower reaction times but preserved accuracy on EF tasks. However, their study did not isolate GAD-specific deficits across different EF domains, as anxiety disorders were analyzed collectively. When GAD was examined separately, their analysis was limited to reaction time and accuracy measures, excluding self-report assessments, which are crucial since perceived cognitive inefficiencies in GAD have been linked to self-esteem and real-world functioning (Henning et al., 2007). Moreover, they did not explore whether GAD disproportionately affects specific EF domains—cognitive flexibility, inhibitory control, working memory—highlighting the need for further investigation into whether GAD-related EF impairments are global or domain-specific.

Given these methodological limitations, a focused systematic review and meta-analysis is needed to isolate EF impairments specifically related to GAD, rather than aggregating findings across anxiety disorders. The current study will examine GAD's impact across distinct EF domains, incorporating both objective and subjective EF measures to determine whether perceived cognitive deficits align with actual impairments. Additionally, it will assess potential moderating factors (e.g., age, GAD measurement, task indices) that may explain discrepancies within the literature. By aggregating the literature in this area, this study aims to refine our understanding of the neurocognitive profile of GAD, ultimately informing diagnostic approaches and targeted interventions.

2. Method

The protocol for this systematic review and meta-analysis was pre-registered in the PROSPERO database (Reference ID: CRD42024591976). This study was conducted following the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins et al., 2024a) and reported in accordance with the PRISMA 2020 guidelines (Page et al., 2021). Data extracted and used in this review are available on the Open Science Framework repository linked to this project (<https://doi.org/10.17605/OSF.IO/7XN6D>).

2.1. Eligibility criteria

Quantitative empirical studies examining the link between GAD diagnosis or symptoms/severity and EFs were included in this review. Qualitative studies and secondary research studies (e.g., reviews, book chapters) were excluded.

Studies exploring group comparisons or correlations were considered for inclusion. Group comparison studies must have included a clinically- or self-diagnosed GAD group and a healthy (no GAD) comparison group. Studies were excluded where the GAD group also had a comorbid psychopathology. This approach was used to isolate the effects of GAD on executive functioning and avoid confounding influences from other psychological disorders. Inclusion of GAD-only samples was determined based on authors' descriptions of participant inclusion and exclusion criteria (e.g., explicit statements that the GAD group did not have

comorbid diagnoses based on clinical interview). In cases where a study included multiple psychopathology and/or comorbidity groups, only those including both a GAD-only group and healthy comparison group were included. Studies lacking a healthy comparison group were excluded. Correlational studies comprising non-clinical samples (e.g., community samples) must have assessed GAD symptoms/severity on a continuum (e.g., symptom measures such as the GAD-7). Inclusion of such studies required that no concurrent psychiatric diagnoses were reported or implied.

Lastly, studies must have administered at least one core EF measure (cognitive flexibility, inhibitory control, working memory; [Diamond, 2013](#); [Miyake et al., 2000](#)). Importantly, EF measures must not have incorporated an affective component (e.g., Flanker task using happy/sad faces) as individuals with GAD have heightened sensitivity to emotional stimuli ([Bui et al., 2017](#)) which may interfere with pure EF performance. Studies only including an EF task using affective stimuli were therefore excluded. Additionally, studies only focusing on cognitive outcomes that are not considered to be a measure of EF (e.g., short-term memory) or other non-cognitive outcomes were also excluded. EF tasks were categorized based on classifications from *A Compendium of Neuropsychological Tests* ([Strauss et al., 2006](#)).

2.2. Information sources

Searches were conducted across multiple databases including PsycINFO, PubMed, Scopus, and Web of Science. The database search was completed on October 12, 2024, with no search restrictions (e.g., date, language). Grey literature was also included by searching ProQuest Conference Papers Index and ProQuest Dissertations and Theses Database. To increase the comprehensiveness of the search and identify any additional potentially relevant articles, forward searching (using Google Scholar's cited by feature) and backward searching (examining reference lists) was conducted. Further, journals relevant to the topic were also manually searched, such as the *Journal of Anxiety Disorders*, *Journal of Affective Disorders*, and *Journal of Abnormal Psychology*. One article was published in Arabic and translated into English using the document translation function on *Google Translate*.

2.3. Search strategy

An initial search string was developed by including key terms identified through relevant articles. A preliminary search was conducted with this search string in Scopus. Additional terms were added to the search string after screening through the first 200 search results (sorted by relevance). The final search string comprised the following key terms:

“generalized anxiety disorder” AND (“executive function*” OR “executive dysfunction*” OR “cognitive control” OR “executive control” OR “central executive” OR inhibit* OR impulsiv* OR “cognitive flexibility” OR “mental flexibility” OR “task switch*” OR “shifting” OR “working memory” OR updating OR “neuropsychological test*” OR “neuropsychological task*”).

2.4. Selection process

The research team reviewed and agreed upon the eligibility criteria prior to engaging in screening to ensure consistency in decisions to include and exclude articles. Covidence software ([Veritas Health Innovation, n.d.](#)) was used to streamline the screening process. After the search process was completed, all search results were imported into Covidence, where duplicates were automatically removed. Duplicates that were missed by Covidence were manually removed. Each article underwent independent review by two authors at both the title/abstract and full-text screening stages. One author screened all articles at each stage, while all other members of the research team screened approximately 20 % of the articles each. Inter-rater reliability was strong across

both title/abstract (Cohen's $\kappa = 0.85$) and full-text screening (Cohen's $\kappa = 0.97$). Discrepancies between reviewers were resolved in team discussions based on the eligibility criteria. In three cases, corresponding authors were contacted for additional information or clarification to determine eligibility; however, no authors responded, so articles were excluded.

2.5. Data collection

Data extraction was conducted manually by the authors working in parallel. A spreadsheet template was designed to systematically record essential study characteristics including participant demographics, study design, measures, and outcomes. To ensure consistency and reliability across authors during the extraction phase, standardized categories and coding instructions were presented to the research team prior to extraction. The template was piloted on a subset of five studies to identify whether any additional data items needed be added to the template. One author was involved in checking the accuracy of extracted data after this stage was completed by the team.

The spreadsheet template included: *study characteristics* (study location, study design), *sample characteristics* (age group, population description, clinical or nonclinical sample, sample size, age, gender), *assessment of GAD* (GAD diagnosis tool and/or GAD symptom/severity measure), *assessment of outcomes* (cognitive domain(s) measured, task(s) administered), *study findings* (results for each outcome), and *methodological quality*.

The methodological quality of each study was evaluated using the Mixed Methods Appraisal Tool (MMAT; [Hong et al., 2018](#)), selected for its ability to assess various study designs including surveys and cross-sectional group comparisons. Each study was assessed across five criteria: i) sampling strategy, ii) representativeness of the sample, iii) appropriateness of measurement tools, iv) potential for nonresponse bias, and v) suitability of statistical analyses. A score ranging from 0 to 5 was attributed to each study, with one point awarded per criterion met. A higher MMAT score reflected stronger methodological rigor. Two authors conducted independent evaluations of each study. Discrepancies were resolved through team discussion.

2.6. Data analysis

2.6.1. Synthesis methods

Meta-analyses were conducted using *Comprehensive Meta-Analysis* (CMA, Version 4) software ([Borenstein et al., 2022](#)). For studies examining correlations between GAD symptoms/severity and EFs, correlation coefficients (r -values, β -weights) and standard errors were entered into CMA. To conduct a comprehensive analysis, the type of data needed to be consistent across studies. Thus, studies reporting group differences (means and standard deviations between GAD and healthy control groups) were converted to standardized mean difference scores following formulas from the Cochrane Handbook. These scores were then transformed into correlation coefficient effect sizes for each outcome ([Higgins et al., 2024b](#)).

Many studies administered multiple tasks to evaluate individual EFs (e.g., n -back and operation span to measure working memory). Following standard meta-analysis procedures, data for each outcome were combined into a single effect size for each study. These aggregated effect sizes were then utilized to determine the overall effect size across studies. The overall effect size was reported as the point estimate (β), where 0.10 represents a small effect size, 0.30 represents a medium effect size, and 0.50 represents a large effect size ([Cohen, 2013](#)). A positive point estimate value suggests that higher GAD symptoms/severity is associated with better EF, whereas a negative point estimate value indicates that higher GAD symptoms/severity is associated with poorer EF.

Data were first examined for potential outliers using CMA's one-study-removed function, which indicates the overall effect size if each

study were to be removed. A study is considered an outlier if its removal alters the interpretation of the findings (i.e., changed statistical significance or size of effect). No outliers were identified.

To gauge the heterogeneity in effect sizes between studies within the analysis, the Q -statistic was reported. The degree of heterogeneity between studies was examined through I^2 , which is interpreted as trivial (< 30 %), moderate (30 %), substantial (50 %), or considerable (75 %) (Deeks et al., 2024). To explore potential sources of variation between studies, moderator analyses were conducted where there was more than a trivial degree of heterogeneity (Borenstein et al., 2021).

Egger's test was used to assess publication bias (Egger et al., 1997), and funnel plots were produced to visualize the distribution of studies reporting positive and negative effect sizes. Additionally, Duval and Tweedie's (2000) trim-and-fill method was used to appraise the potential impact of publication bias by imputing missing studies to create a symmetrical funnel plot, and calculating an effect size adjusted for bias. If the interpretation of findings (e.g., size of effect, statistical significance) is consistent between the original and adjusted effect sizes, publications bias is considered to be trivial; however, if the interpretation of findings differs, the potential impact of bias is considered to be substantial (Borenstein et al., 2021).

2.6.2. Planned analyses

To account for heterogeneity between study designs, populations, and outcome measures, random-effects meta-analyses were conducted, with an alpha level of 0.05 to determine statistical significance. A meta-analysis was conducted to examine the relationship between GAD symptoms/severity and overall EF. Separate analyses were also conducted for each EF domain (i.e., cognitive flexibility, inhibitory control, working memory).

To provide deeper insight into the relationship between GAD and EFs, planned moderator analyses were also conducted for the following variables: i) *sample type* (clinical [sample formally diagnosed with GAD], non-clinical [sample not formally diagnosed with GAD e.g., community/college sample]), ii) *age group* (youths [up to 19 years], young/middle-aged adults [20 to 59 years], older adults [60 years and older]),¹ iii) *task index* (accuracy, reaction time, self-report), iv) *GAD operationalization* (GAD diagnosis [comparing group diagnosed with GAD and group not diagnosed with GAD], GAD symptoms/severity [examining GAD as a continuum]), and v) *GAD measurement* (clinical interview, GAD scale).²

2.6.3. Sensitivity analyses

To assess whether the findings are robust against key methodological and analytical decisions made throughout the review, sensitivity analyses were conducted for model type (fixed- versus random-effects model), study design (correlational versus group comparison studies), methodological quality as assessed by the MMAT (low [MMAT score of 0 to 2] versus high [MMAT score of 3 to 5]). Across all studies and outcomes, the mean effect size was comparable for fixed-effects ($k = 32$, $\beta = -0.19$, $p < .001$) and random-effects models ($k = 32$, $\beta = -0.24$, $p < .001$). Similarly, both correlational ($k = 10$, $\beta = -0.21$, $p = .005$) and group comparison studies ($k = 22$, $\beta = -0.26$, $p < .001$) yielded similar effect sizes. Lastly, studies appraised as having lower methodological

quality ($k = 15$, $\beta = -0.22$, $p < .001$) reported similar effect sizes to studies appraised as having higher methodological quality ($k = 17$, $\beta = -0.27$, $p = .001$). The findings therefore appeared to be robust against key methodological and analytical factors chosen and included in the current meta-analysis as the findings and interpretations remained consistent across different assumptions made.

3. Results

3.1. Search results

From the 3329 articles screened, 32 studies met all eligibility criteria for inclusion in the current systematic review and meta-analysis. Fig. 1 presents the PRISMA flow diagram depicting the study selection process throughout the review. See Table 1 for summary of study characteristics and findings.

3.2. Characteristics of included studies

3.2.1. Study characteristics

The geographic distribution of studies spanned multiple continents, with the majority conducted in North America (41 %) and Europe (22 %), followed by Asia (31 %), and fewer studies being conducted in Africa (3 %) and South America (3 %). The studies were relatively even in methodological approach, with slightly fewer correlational designs (34 %) than group comparison designs (66 %).

There was an approximately equivalent proportion of studies appraised as having high methodological quality (53 %; MMAT score of 3 to 5) and low methodological quality (47 %; MMAT score of 0 to 2). For group comparison studies, the most common MMAT criteria that were not met related to sample representativeness and accounting for confounding variables. Similarly, for correlational studies sample representativeness was a common issue, as well as high risk of non-response bias.

3.2.2. Participant characteristics

A total of 13,084 participants were involved across all 32 studies included in this review. There was high variability in the sample size across studies, with the number of participants ranging from 30 to 4240 ($M = 408.88$; $SD = 887.62$). Correlational studies generally had larger sample sizes ($M = 978.91$, $SD = 1348.21$, range = 93 to 4240) than group comparison studies ($M = 110.29$, $SD = 200.08$, range = 30 to 970). Within group comparison studies, the mean sample size for GAD groups was smaller ($M = 27.19$, $SD = 13.08$) than for healthy control groups ($M = 83.10$, $SD = 192.14$).

The majority of studies focused on young and middle-aged adult participants (78 %), with fewer studies focusing on youths (13 %) and older adults (9 %). Overall, the average age of participants was 32.45 years ($SD = 6.29$), and the gender distribution, on average, was more prominently female (62.9 %) than male (37.1 %). More studies involved clinical samples (59 %) than non-clinical samples (e.g., general population, college students; 41 %).

3.2.3. Measurement characteristics

Most studies operationalized GAD using a formal diagnosis (66 %), while fewer relied on symptom/severity measures (34 %). Similarly, GAD was mostly measured using clinical interviews (66 %) compared to GAD scales (34 %). Different facets of EF were also measured across studies, with cognitive flexibility being the most frequently assessed domain (50 %), followed by working memory (38 %), and inhibitory control (34 %). Multiple EF domains were assessed across 22 % of studies.

3.3. Impact of GAD on executive functioning

Table 2 presents a summary of mean effect sizes and summary effect

¹ Age groups categorized based on United Nations' definitions of children and adolescents (up to 19 yrs) and older persons (60+ yrs).

² Although conceptually related, the moderators *Sample Type* (clinical vs non-clinical), *GAD Operationalization* (diagnosis vs symptoms/severity), and *GAD Measurement* (clinical interview vs GAD scale) were coded separately due to meaningful differences in study design and analytic focus. For example, some studies used clinical interviews to assess symptom severity rather than diagnosis, while others examined GAD symptom/severity as a continuum within clinical samples without comparing to controls (not diagnosis-based), necessitating distinction between sample type, how GAD was operationalized, and how it was measured.

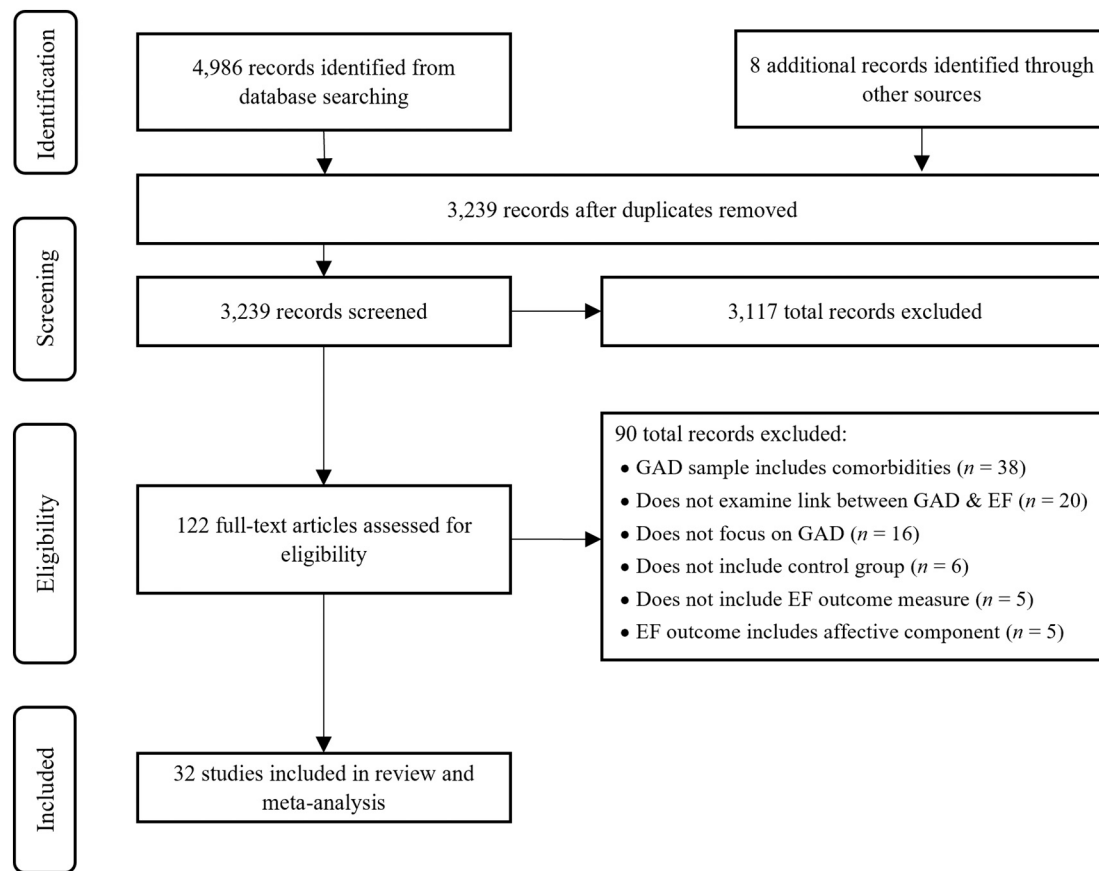


Fig. 1. PRISMA flow chart.

sizes for overall EF and individual domains. Across all EF outcomes, a significant, negative, small mean effect size was obtained across 32 studies, $\beta = -0.24$, $CI_{95}[-0.34, -0.15]$, $p < .001$ (see Fig. 2 for forest plot stratified by EF domain). This indicates that GAD is associated with poorer EF. The likely impact of publication bias on this finding was trivial. Egger's test indicated that the funnel plot was not significantly asymmetrical ($\beta_{\text{Egger's}} = -2.33$, $SE = 1.47$, $p = .122$); however, the trim-and-fill analysis imputed 9 studies of low and high precision with negative effect sizes to balance the funnel plot. Whilst this yielded a slightly stronger negative mean effect size adjusted for bias ($\beta_{\text{adjusted}} = -0.29$, $CI_{95}[-0.37, -0.20]$), the interpretation of findings (significant, small, negative effect) remained unchanged (see Fig. 3).

Significant, considerable heterogeneity of effect sizes was evident across studies, $Q(31) = 996.22$, $p < .001$, $I^2 = 96.89\%$, which was further explored through moderator analyses. Sample type was not a significant moderator, $Q(1) = 0.49$, $p = .486$, indicating that similar effect sizes were obtained in studies involving clinical and non-clinical samples. Age group was also a nonsignificant moderator, $Q(2) = 0.81$, $p = .666$. Notably, however, studies involving youths and young/middle-aged adults yielded significant effect sizes but not studies involving older adults (see Table 2 for summary of effect sizes). Task index was a significant moderator, $Q(2) = 28.15$, $p < .001$, with self-report measures eliciting stronger negative effect sizes than accuracy-based and reaction time-based measures. GAD operationalization (diagnosis/symptoms) and GAD measurement (interview/scale) were not significant moderators ($ps \geq 0.725$).

3.3.1. Cognitive flexibility

Cognitive flexibility was assessed across 19 studies using measures such as the Wisconsin Card Sorting Task, fluency tasks (e.g., Ruff Figural Fluency Test), and self-report measures such as the Shifting subscale of

the Attention Control Scale. These studies yielded a significant, moderate, negative mean effect size, $\beta = -0.30$, $CI_{95}[-0.40, -0.19]$, $p < .001$, indicating that GAD is associated with poorer cognitive flexibility. There was trivial likely impact of publication bias (nonsignificant funnel plot asymmetry, $\beta_{\text{Egger's}} = -1.88$, $SE = 1.94$, $p = .344$, and no imputed studies from the trim-and-fill analysis). However, significant, considerable heterogeneity was evident across studies, $Q(18) = 415.89$, $p < .001$, $I^2 = 95.67\%$, warranting exploration of potential moderators.

Sample type (clinical/non-clinical) and age group were not significant moderators ($ps \geq 0.247$); though, only studies involving older adults did not yield significant effect sizes. Task index, however, was a significant moderator, $Q(2) = 10.53$, $p < .001$, revealing that GAD was most strongly associated with poorer self-reported cognitive flexibility, followed by poorer accuracy (but not associated with reaction time). GAD operationalization (diagnosis/symptoms) and GAD measurement (interview/scale) were not significant moderators ($ps \geq 0.557$).

3.3.2. Inhibitory control

Inhibitory control was assessed by 11 studies through measures such as the Flanker task, Stroop task, and Stop Signal Task. The mean effect size across these studies was nonsignificant, $\beta = -0.19$, $CI_{95}[-0.41, 0.04]$, $p = .099$, indicating no relationship between GAD and inhibitory control. Notably, whilst Egger's test indicated no significant asymmetry in the funnel plot ($\beta_{\text{Egger's}} = -1.65$, $SE = 3.44$, $p = .643$), the trim-and-fill analysis imputed 4 studies of low and moderate precision with negative effect sizes to balance the funnel plot (see Fig. 4). The mean effect size adjusted for bias ($\beta_{\text{adjusted}} = -0.32$, $CI_{95}[-0.49, -0.10]$) changed the interpretation of the findings as the mean effect size became significant, indicating that the likely impact of publication bias on the findings is substantial. The findings from this analysis must therefore be taken with caution.

Table 1

Summary of study characteristics and findings for studies included in review.

Study characteristics		Assessment of GAD		Findings and methodological quality	
Authors, study location	Sample description	Assessment type (diagnosis/symptoms)	Measure(s)	Key findings	MMAT (/5)
Abasi et al. (2017), Iran	524 adults from general population ($M_{age} = 35.24$, $SD = 10.71$, range = 20–60 yrs; 53.6 % male)	Questionnaire (symptoms)	GAD-7	• GAD-7 ↓ self-reported CF score	3
Abushalmaq et al. (2021), Egypt	20 adults with GAD, 24 HC adults ($M_{age} = 43.11$, $SD = 6.15$, range = 30–60 yrs; 56.8 % male)	Clinical interview (diagnosis)	Mini International Neuropsychiatric Interview (MINI)	• GAD group ↓ WM maintenance (accuracy) compared to HC group • ns group differences in WM duration and load (accuracy)	3
Airaksinen et al. (2005), Sweden	7 adults with GAD, 175 HC adults ($M_{age} = 42.80$, $SD = 12.05$, range = 20–64 yrs; 50.0 % male)	Clinical interview (diagnosis)	Schedules for Clinical Assessments in Neuropsychiatry (SCAN)	• GAD group ↓ CF (slower RT) compared to HC group • ns group differences in CF (accuracy)	3
Armstrong et al. (2011), USA	29 adults with GAD, 29 HC adults ($M_{age} = 38.88$, $SD = 10.64$, range = 18+ yrs; 49.0 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-IV (SCID-I)	• GAD group ↓ self-reported CF compared to HC group	3
Baumel et al. (2022), USA	50 adolescents with GAD, 50 HC adolescents ($M_{age} = 14.80$, $SD = 1.60$, range = 12–17 yrs; 24.0 % male)	Questionnaire (diagnosis)	Pediatric Anxiety Rating Scale (PARS)	• GAD group ↓ self-reported CF, IN, and WM scores compared to HC group	3
Bredemeier and Berenbaum (2013), USA	198 college students ($M_{age} = 19.60$, $SD = 1.70$, range = 18–26 yrs; 44.0 % male)	Clinical interview (symptoms)	Structured Clinical Interview for DSM-IV Disorders, Nonpatient Edition (SCID-NP)	• Dimensional GAD score ↓ WM accuracy (2-back) • GAD ns association WM accuracy (Operation Span task)	2
Cavanagh et al. (2017), USA	39 adults with GAD, 52 HC adults ($M_{age} = 25.93$, $SD = 9.27$, range = 18+ yrs; 10.0 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-IV (SCID-I)	• GAD group ↓ IN (slower RT on post-error trials) compared to HC group • ns group differences in IN task accuracy and RT (correct and incorrect trials)	3
Coussement et al. (2022), Belgium	30 adults with GAD, 29 HC adults ($M_{age} = 29.37$, $SD = 9.98$, range = 18+ yrs; 22.5 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-V (SCID-I)	• GAD group ↓ IN (slower RT) compared to HC group	3
Dorenkamp et al. (2023), USA	93 older adults from a community sample ($M_{age} = 70.79$, $SD = 7.97$, range = 55–89 yrs; 34.4 % male)	Questionnaire (symptoms)	GAD-7	• ns association GAD-7 with CF (Trail Making Task RT), IN (Stroop RT), WM (Digit Span accuracy)	2
García-Martín et al. (2021), Colombia	15 adults with GAD, 90 HC adults ($M_{age} = 22.94$, $SD = 4.62$, range = 18–38 yrs; 35.2 % male)	Questionnaire (symptoms)	GAD-7	• GAD ↓ IN accuracy (Stroop word, color, word-color) compared to HC • ns group difference IN (Stroop interference accuracy)	2
Golajj et al. (2014), Iran	17 adults with GAD, 15 HC adults ($M_{age} = 40.26$; 20.0 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-V (SCID-I)	• ns group differences in WM accuracy (Letter Number Sequencing)	2
Gualtieri and Morgan (2008), USA	63 adults with GAD, 907 HC adults ($M_{age} = 40.26$, range = 18–65 yrs; 41.2 % male)	Clinical interview (diagnosis)	Psychiatric screening using DSM-IV-TR criteria	• GAD ↓ CF (accuracy) compared to HC • GAD ↓ IN (slower RT) compared to HC	2
Hazlett-Stevens (2000), USA [Thesis]	26 college students with GAD, 23 HC college students ($M_{age} = 19.98$, $SD = 1.55$; 10.2 % male)	Questionnaire (symptoms)	Self-reported symptoms against DSM-IV diagnostic criteria	• GAD ↓ self-reported CF (Cognitive Flexibility Scale) and CF accuracy (Design Fluency Task) compared to HC	1
Hu et al. (2020), China	25 adults with GAD, 25 HC adults ($M_{age} = 28.85$; 44.0 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-V (SCID-I)	• GAD ↓ IN accuracy (antisaccade task)	3
John (2006), Canada [Thesis]	42 children with GAD, 34 HC children ($M_{age} = 10.08$, $SD = 1.23$, range = 7–12 yrs; 61.8 % male)	Clinical interview (diagnosis)	Anxiety Disorders Interview Schedule for DSM-IV (ADIS)	• GAD ↓ WM accuracy (CHIPASAT) – some indices only • GAD ↓ WM accuracy (Backward Digit Span) – some indices only	2
Kahraman et al. (2022), Turkey	275 college students ($M_{age} = 20.55$, $SD = 2.15$, range = 18+ yrs; 25.5 % male)	Questionnaire (symptoms)	GAD-7	• GAD ↓ self-reported CF (Cognitive Control & Flexibility Questionnaire)	3
Kalanthorff et al. (2017), Israel	26 adults with GAD, 35 HC adults ($M_{age} = 27.37$, $SD = 7.61$, range = 18–64 yrs; 24.6 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-V (SCID-I)	• ns association IN RT (Objective Interference and Stop-Signal Task)	2
Lea et al. (2018), USA	21 college students with GAD, 20 HC college students ($M_{age} = 18.97$, $SD = 0.87$; 26.9 % male)	Clinical interview (diagnosis)	Semi-structured clinical interview by clinical psychologist using DSM-V	• ns group difference WM accuracy (phonological & visuospatial WM tasks)	3

(continued on next page)

Table 1 (continued)

Study characteristics		Assessment of GAD		Findings and methodological quality	
Authors, study location	Sample description	Assessment type (diagnosis/symptoms)	Measure(s)	Key findings	MMAT (/5)
Li and Fan (2022), China	567 college students ($M_{age} = 21.40$, $SD = 2.25$, range = 18+ yrs; 42.5 % male)	Questionnaire (symptoms)	GAD-7	• GAD-7 ↓ self-reported CF (Attention Shifting subscale of Attentional Control Scale)	3
Luo et al. (2022), China	37 adults with GAD, 74 HC adults ($M_{age} = 36.45$, $SD = 6.58$, range = 18+ yrs; 41.4 % male)	Clinical interview (diagnosis)	Diagnosis from psychiatrist; HCs confirmed using Structured Clinical Interview for DSM-V Disorders, Nonpatient Edition (SCID-NP)	• GAD ↓ WM accuracy (Spatial WM task) compared to HC – some indices only • GAD ↓ CF accuracy (Intra-Extra Dimensional Set Shift) compared to HC	2
Malivoire et al. (2019), Canada	274 adults from the general population ($M_{age} = 40.30$, $SD = 12.90$, range = 18+ yrs; 33.0 % male)	Questionnaire (symptoms)	GAD-Q-IV	• GAD-Q-IV ↓ self-reported cognitive flexibility score	3
Mantella et al. (2007), USA	19 older adults with GAD, 40 HC older adults ($M_{age} = 70.05$, $SD = 6.85$, range = 60+ yrs; 44.1 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-V (SCID-I)	• GAD ↓ CF (slower RT on Trail Making Test B) compared to HC • ns group difference CF RT (Trail Making Test A and Ratio)	2
Moghadasin and Dibajnia (2021), Iran	45 adults with GAD, 50 HC adults ($M_{age} = 29.91$, $SD = 8.00$, range = 18–65 yrs; 46.3 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for DSM-V (SCID-5-CV)	• GAD ↓ WM accuracy (Working Memory Index) compared to HC	2
Pawluk and Koerner (2013), Canada	140 college students ($M_{age} = 19.59$, $SD = 2.94$, range = 18+ yrs; 17.9 % male)	Questionnaire (symptoms)	GAD-Q-IV	• GAD-Q-IV ↓ self-reported IN (lower functional impulsivity) • ns association GAD-Q-IV with self-report IN (impulsiveness and dysfunctional impulsivity)	3
Sportel et al. (2011), Netherlands	1806 school students ($M_{age} = 13.60$, $SD = 0.66$; 45.0 % male)	Questionnaire (symptoms)	Revised Child Anxiety and Depression Scale (RCADS)	• GAD ↓ self-reported CF (Attentional Control subscale)	2
Stefanopoulou et al. (2014), UK	17 adults with GAD, 17 HC adults ($M_{age} = 32–55$ yrs; 23.5 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for the DSM-IV (SCID-I)	• GAD ↓ WM updating (longer RT on 2-back and 3-back) compared to HC • ns group difference WM updating (RT on 0-back and 1-back, accuracy on 0- to 3-back) compared to HC	3
Szabó et al. (2013), Hungary	21 adults with GAD, 30 HC adults	Clinical interview (diagnosis)	Structured Clinical Interview for DSM-IV Axis I Disorders - Clinician Version (SCID-CV)	• ns group difference CF accuracy (Wisconsin Card Sorting Task)	2
Tang et al. (2022), China	30 adults with GAD, 30 HC adults ($M_{age} = 32.50$, range = 18–60 yrs; 46.7 % male)	Clinical interview (diagnosis)	Clinical diagnosis of GAD based on GAD diagnostic criteria of DSM-V	• GAD ↓ WM accuracy and RT (1-back) • GAD ↓ WM accuracy (2-back) • ns group difference IN accuracy and RT (Go/No-Go) and WM RT (2-back)	3
Tempesta et al. (2013), Italy	22 adults with GAD, 31 HC adults ($M_{age} = 31.55$, $SD = 7.50$, range = 20–35 yrs; 34.0 % male)	Clinical interview (diagnosis)	Structured Clinical Interview for DSM-IV-Patient Edition (SCID-P)	• GAD ↓ CF accuracy (Wisconsin Card Sorting Task) compared to HC – some indices only • ns group differences WM accuracy (Digit Span, Corsi Blocks)	2
Unterrainer et al. (2018), Germany	4240 adults from the general population ($M_{age} = 60.00$, $SD = 10.80$, range = 40–80 yrs; 51.2 % male)	Questionnaire (symptoms)	GAD-7 (2-item version)	• GAD ↓ EF accuracy (Tower of London) compared to HC	3
Yang et al. (2015), China	15 adults with GAD, 15 HC adults ($M_{age} = 30.25$, $SD = 5.50$, range = 18+ yrs; 46.7 % male)	Clinical interview (diagnosis)	Clinical diagnosis of GAD based on GAD diagnostic criteria of DSM-V	GAD ↓ EF (slower RT in Dual-Feature Delayed Matching Paradigm) compared to HC – task-relevant mismatch condition • ns group difference EF accuracy (all conditions) and RT (match, task-irrelevant and conjunction mismatch conditions)	2
Zainal and Newman (2021), USA	2581 adults from the general population ($M_{age} = 46.00$, $SD = 11.40$, range = 24–74 yrs; 45.3 % male)	Clinical interview (symptoms)	Composite International Diagnostic Interview-Short Form (CIDI-SF)	• ns association GAD with CF accuracy and RT or IN accuracy and RT (Stop-and-Go Switch Task)	4

Note. ↓ = GAD associated with poorer EF outcome, ↑ = GAD associated with better EF outcome, ns = nonsignificant association; GAD = Generalized Anxiety Disorder, HC = healthy control; GAD-7 = General Anxiety Disorder-7 item scale; RT = reaction time; CF = cognitive flexibility (shifting/switching), IN = inhibitory control (inhibition), WM (updating) = working memory.

Due to the significant, considerable heterogeneity in effect sizes across studies, $Q(10) = 456.50$, $p < .001$, $I^2 = 97.81$ %, moderator analyses were conducted to further explore potential sources of variance. Neither sample type (clinical/non-clinical sample) nor age group were significant moderators ($ps \geq 0.106$), with all studies yielding nonsignificant effect sizes except for studies involving youths. Whilst task

index was also not a significant moderator, $Q(2) = 3.43$, $p = .180$, GAD was associated with lower self-reported inhibitory control, but not reaction time and accuracy on inhibitory control tasks. GAD operationalization and GAD measurement were not significant moderators ($ps \geq 0.395$), indicating that studies examining GAD diagnosis/symptoms and studies utilizing clinical interviews/GAD scales yielded comparable

Table 2

Summary and moderator effect sizes for impact of GAD on executive functioning.

Moderator/categories	Executive functioning domains							
	Overall EF		Cognitive flexibility		Inhibitory control		Working memory	
	β	<i>k</i>	β	<i>k</i>	β	<i>k</i>	β	<i>k</i>
Overall effect size	−0.24***	32	−0.30***	19	−0.19	11	−0.20***	12
Sample type								
Clinical	−0.27***	19	−0.37***	9	−0.24	7	−0.29***	9
Non-clinical	−0.20**	13	−0.23**	10	−0.08	4	−0.01	3
Age group								
Older adult	−0.15	3	−0.17	2	0.04	1	0.14	1
Young/middle-aged adults	−0.26***	25	−0.32***	14	−0.21	9	−0.22***	8
Youth	−0.27***	4	−0.30**	3	−0.25**	1	−0.25*	3
Task index								
Accuracy	−0.18***	29	−0.26*	10	−0.20	5	−0.17***	11
Reaction time ^a	−0.09**	14	−0.10	4	−0.07	7	−0.26**	2
Self-report	−0.36***	12	−0.38***	8	−0.21**	2	−0.47***	1
GAD operationalization								
GAD diagnosis	−0.26***	21	−0.32***	11	−0.24	7	−0.27***	10
GAD symptoms	−0.22**	11	−0.26***	8	−0.08	4	−0.10**	2
GAD measurement								
Clinical interview	−0.23***	21	−0.27**	11	−0.20	7	−0.24***	9
GAD scale	−0.26***	11	−0.33***	8	−0.17	4	−0.14	3

Note. *k* represents the number of studies included in each category, β represents the effect size (0.10 = small, 0.30 = moderate, 0.50 = strong).

* $p \leq .05$.

** $p \leq .01$.

*** $p \leq .001$.

^a Lower scores on reaction time indicate poorer performance (i.e., slower reaction time).

effect sizes.

3.3.3. Working memory

Working memory was measured in 12 studies through tasks such as the *N*-back, Digit Span, and Letter-Number Sequencing. Overall, these studies revealed a significant, small, negative effect size, $\beta = -0.20$, $CI_{95}[-0.31, -0.09]$, $p < .001$, indicating that GAD is associated with poorer working memory. The likely impact of publication bias on this finding was trivial. The funnel plot displayed no significant asymmetry ($\beta_{\text{Egger's}} = 0.83$, $SE = 2.01$, $p = .688$) and the trim-and-fill analysis imputed no studies.

Moderator analyses were conducted to explore the significant substantial heterogeneity across study effect sizes, $Q(11) = 35.03$, $p < .001$, $I^2 = 68.59\%$. Sample type was a significant moderator, $Q(1) = 12.37$, $p < .001$, where only studies involving clinical samples (but not non-clinical samples) yielded significant effect sizes. Similarly, age group was a significant moderator, $Q(1) = 10.27$, $p = .006$, with studies involving youths and young/middle-aged adults yielding significant effect sizes (but not studies involving older adults). This suggests that GAD is associated with poorer working memory in clinical samples and younger populations. Task index was also a significant moderator, $Q(2) = 10.22$, $p = .006$. Significant, negative effect sizes were obtained for each task index; however, the effect size was stronger for self-report measures (moderate effect), compared to accuracy and reaction time measures (small effect). GAD operationalization and measurement were not significant moderators ($ps \geq 0.444$).

4. Discussion

This systematic review and meta-analysis sought to clarify the relationship between GAD and EFs by synthesizing findings across 32 studies. Given the mixed reports pertaining to EF deficits in GAD, this study examined the link between GAD and each core EF domain, while also exploring potential moderators including sample type, age group, EF task index, GAD operationalization, and GAD measurement. The findings from this meta-analysis extend upon previous reviews in this area and yield important theoretical and clinical implications for assessment and intervention strategies to support individuals with GAD.

4.1. Link between GAD and executive functioning

Overall, our findings partially support and extend upon the findings of [Majeed et al. \(2023\)](#) who found a negative effect of GAD on reaction time, with no significant difference in accuracy between GAD and control groups. The authors explain this in relation to the attentional control theory which posits that individuals with high anxiety allocate greater cognitive resources to maintain task accuracy, often at the expense of processing efficiency ([Eysenck et al., 2007](#); [Leonard and Abramovitch, 2019](#)). In contrast, our meta-analysis revealed that individuals with GAD exhibited poorer performance on both reaction time and accuracy across EF tasks overall. This discrepancy may be attributed to differences in statistical power. The current meta-analysis included 29 studies in the accuracy analysis and 14 for reaction time, whereas [Majeed et al. \(2023\)](#) included nine studies for each task index. Hence, the current study may provide a more robust estimate of GAD-related EF impairments. Our finding therefore suggests that the attentional control theory may not fully account for the neuropsychological profile observed in individuals with GAD. A potential explanation is that the heightened and pervasive nature of worry characteristic of GAD depletes cognitive resources more broadly, resulting in widespread executive dysfunction rather than a compensatory focus on maintaining task accuracy. For example, individuals with GAD may experience constant internal distraction from intrusive worry, which taxes working memory and inhibits flexible shifting between task demands, leading to both slower performance and reduced accuracy—particularly on tasks requiring sustained attention and cognitive flexibility under pressure.

Extending upon prior reviews in this area, this study provided a synthesis of the specific impact of GAD across each EF domain. Significant, negative effects of GAD were observed for cognitive flexibility and working memory, but not inhibitory control. The lack of an observed effect on inhibitory control may reflect that GAD selectively impacts certain aspects of this domain. For instance, [Hallion et al. \(2017\)](#) found that while individuals with GAD performed worse than controls on the Stroop task, their performance on the Go/No-Go task was comparable to that of healthy controls. The authors explained that the Stroop task requires the inhibition of a cognitive process (e.g., suppressing the automatic tendency to read a word), whereas the Go/No-Go task assesses the inhibition of motor responses. Given that cognitive inhibition aligns

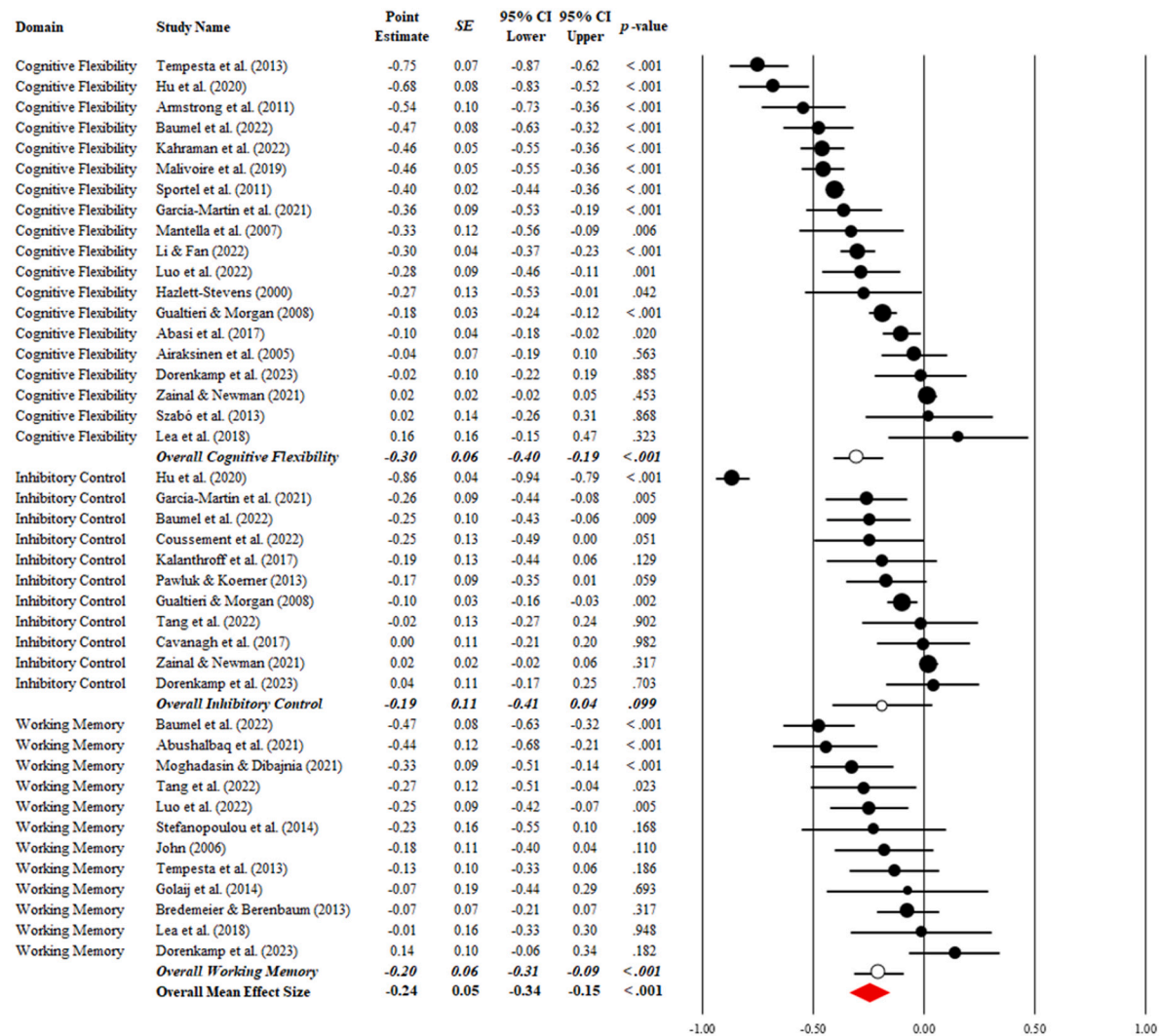


Fig. 2. Forest plot summarizing effect sizes of studies, stratified by EF domain.

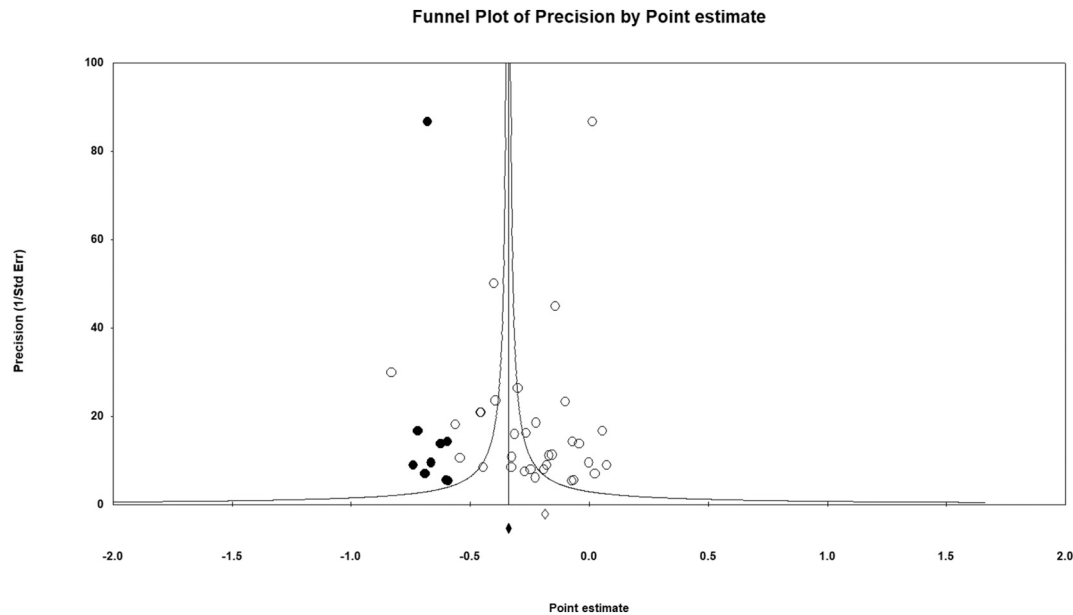


Fig. 3. Funnel plot depicting potential impact of publication bias for overall EF analysis.

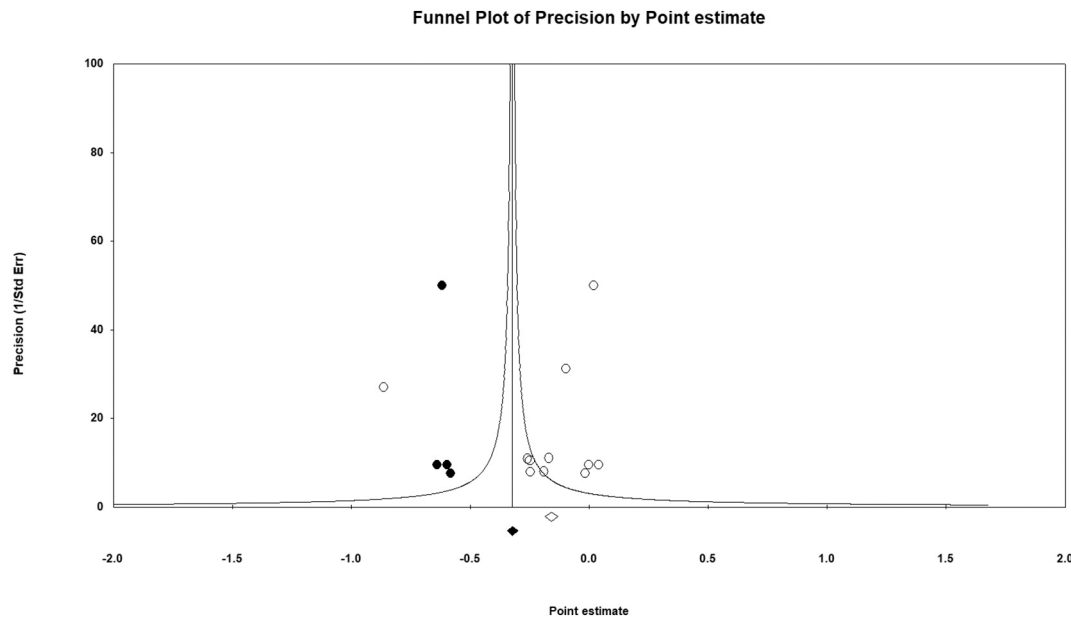


Fig. 4. Funnel plot depicting potential impact of publication bias for inhibitory control analysis.

more closely with selective attention (a cognitive process heavily impacted by anxiety; Richards et al., 2014), this may explain why GAD appears to affect some but not all inhibitory control tasks. Notably, however, the current meta-analysis indicated substantial (potential) impact of publication bias for the analysis surrounding inhibitory control; if studies potentially missing due to publication bias were accounted for, a significant negative impact of GAD on this domain would be observed. This finding should therefore be taken and interpreted with caution. Further research is warranted to explore whether (and if so, how) GAD might impact different facets of inhibitory control.

In contrast, GAD was associated with poorer accuracy on cognitive flexibility tasks and global impairment in working memory (i.e., reduced accuracy and slower reaction times). These findings challenge attentional control theory, which posits reduced processing efficiency (slower responses) but preserved performance effectiveness (accuracy) under anxiety (Eysenck et al., 2007). The negative association between GAD and working memory is well-documented (Abushalbak et al., 2021; Grant and Chamberlain, 2023; MacNamara and Proudfoot, 2014; Park et al., 2015), and is commonly attributed to excessive worry depleting limited cognitive resources. Individuals with GAD often experience persistent, verbal-linguistic worry that occupies working memory capacity and impairs the maintenance and manipulation of task-relevant information, particularly under cognitive load (Leigh and Hirsch, 2011; Toh and Vasey, 2017).

The accuracy deficits in cognitive flexibility, however, are less frequently reported and warrant further consideration. One explanation is that anxiety-related attentional biases and perseverative worry impair the ability to shift between mental sets, which is a core feature of cognitive flexibility (Park and Moghaddam, 2017). Given that cognitive flexibility depends on the ability to update and manipulate information in working memory (Diamond, 2013), it is likely that impairments in working memory contribute to difficulties in cognitive flexibility. This relationship is supported by neurobiological evidence indicating that both working memory and cognitive flexibility engage the dorsolateral prefrontal cortex—a region particularly susceptible to anxiety-related disruption (White et al., 2023). When this neural system is compromised, individuals may struggle to reconfigure cognitive strategies or maintain task-relevant rules, reducing accuracy despite preserved response speed. This may explain the observed mismatch between accuracy and reaction time in flexibility tasks. Critically, however, it should be noted that only 3 studies assessed reaction time on cognitive

flexibility tasks, warranting further research into the relationship between GAD and cognitive flexibility, as well as potential mediating relationships with working memory.

Lastly, another novel finding of this study was that GAD had a negative effect on self-report measures across each EF domain, with these effect sizes exceeding those of objective measures (both accuracy and reaction time indices). This suggests that subjective cognitive deficits may be a crucial component of GAD-related dysfunction and aligns with prior research indicating that anxiety amplifies perceived cognitive impairment, even when objective performance may be relatively intact (Hill et al., 2016). This highlights the need to interpret self-reported EF impairments with caution, as they may reflect anxiety-related biases rather than actual deficits in cognitive functioning. However, given that perceived cognitive ability can influence actual performance, self-esteem, and real-world functioning (e.g., Henning et al., 2007), interventions targeting confidence and self-perception may be beneficial for improving both cognitive functioning and overall well-being in individuals with GAD. Additionally, incorporating cognitive training exercises targeting working memory and cognitive flexibility may help elevate both subjective and objective indices of EF, ultimately addressing both the perceived and actual cognitive challenges faced by individuals with GAD.

4.2. Moderators of the link between GAD and executive functioning

Sample type, GAD operationalization, age, and GAD measurement were explored as potential moderators of the association between GAD and EFs, though findings were mixed across different EF domains. Despite sample type being a significant moderator for working memory outcomes only, studies involving clinical GAD samples more consistently yielded significant (and larger) negative effect sizes than studies involving non-clinical (e.g., community) samples. Similarly, whilst GAD operationalization was not a significant moderator, studies focusing on GAD diagnoses yielded slightly stronger effect sizes than studies focusing on GAD symptoms/severity. This pattern of findings is not surprising given that clinically diagnosed samples likely represent individuals with more severe or impairing forms of GAD, which may have a stronger impact on EF. Diagnostic status may also reflect greater chronicity which is associated with poorer cognitive performance. Categorical diagnoses may also capture more distinct group differences than symptom/severity-based dimensional measures, contributing to

the slightly stronger effect sizes observed in diagnosis-focused studies.

Although age group was not a consistent moderator across EF domains, effect sizes for studies involving youths and young/middle-aged adults were consistently larger and more robust than those for older adults, suggesting that GAD may have a greater impact on EF earlier in the lifespan. This aligns with findings from [Gulpers et al. \(2022\)](#), who reported poorer EF among youths with higher GAD symptoms, but contrasts with [Majeed et al. \(2023\)](#), who found no moderating effect of age in a broader anxiety sample. This discrepancy may reflect differences in diagnostic specificity, as [Majeed et al. \(2023\)](#) included multiple anxiety disorders (e.g., panic disorders, specific phobia) that may be less sensitive to developmental variation. These findings support the idea that developmental factors may shape the relationship between GAD specifically (rather than anxiety generally) and EF, particularly in cognitive flexibility and working memory—domains vulnerable during critical periods of cognitive maturation (e.g., [Cowan, 2014](#)). The significant association with inhibitory control in youths, although preliminary, may reflect their heightened vulnerability due to ongoing prefrontal cortex development ([Casey et al., 2008](#)), with potential implications for academic outcomes and emotion regulation ([Rabner et al., 2024](#)).

In contrast, the nonsignificant associations observed in older adults could reflect normative age-related cognitive decline which may obscure GAD-related impairments, particularly in prefrontal regions involved in EF ([West, 1996](#); [Zelazo et al., 2004](#)). Further, in completing cognitive tasks, some older adults may draw upon cognitive reserve or compensatory strategies that buffer against the cognitive impact of anxiety ([Stern, 2002](#)). Such adaptive mechanisms may help maintain task performance despite underlying anxiety symptoms, thereby reducing the likelihood of detecting significant associations between GAD and EF. In addition, GAD in older adults often presents with fewer and qualitatively different symptoms than in younger individuals, with reduced cognitive hyperarousal and a greater focus on somatic concerns ([Beekman et al., 2000](#); [Miloyan et al., 2014](#)). These symptoms can overlap with common features of normal aging (e.g., fatigue, sleep disturbance, cognitive slowing) which may obscure the detection of GAD and its specific effects on EF. This pattern highlights the need for more age-sensitive assessments of GAD that can reliably distinguish pathological anxiety from normative age-related changes. Critically, however, as the number of studies involving youths and older adult populations were limited in the current review, these findings should be taken with caution. Further research is therefore warranted to clarify these potential developmental differences, in order to better understand and support individuals with GAD across the lifespan.

Lastly, GAD measurement was not a significant moderator for any EF domain, suggesting that effect sizes were similar for studies measuring GAD using clinical interviews and studies administering GAD scales. Both methods yielded significant effect sizes for cognitive flexibility and no significant effect sizes for inhibitory control. Notably, however, only studies administering clinical interviews (but not studies using GAD scales) yielded significant effect sizes for working memory. This may be due to clinical interviews capturing a more comprehensive and clinically relevant picture of GAD severity, including functional impairment and symptom chronicity, which are more likely to be associated with working memory deficits (e.g., [Lukasik et al., 2019](#)). In contrast, self-report scales may be more sensitive to transient or subclinical anxiety symptoms that exert less consistent effects on cognitive performance. As working memory is particularly sensitive to sustained and high-intensity worry (e.g., [Held et al., 2020](#)), studies using interviews may have better identified individuals experiencing the degree of impairment necessary to detect significant effects. However, due to the limited number of studies examining GAD (measured by scale) and working memory, further research is warranted to better understand whether differences in measurement methods meaningfully influence the detection of GAD-related cognitive impairments.

4.3. Limitations and future directions

The findings of this meta-analysis offer important insights into the relationship between GAD and EFs; however, several methodological limitations must be considered when interpreting the results. This meta-analysis was limited to studies comprising participants without comorbid disorders, despite high rates of comorbidity in GAD—particularly with depression and other anxiety disorders ([Kessler et al., 2005](#); [Mennin et al., 2008](#)). This was done to isolate the specific impact of GAD on EFs. However, since comorbid conditions may amplify cognitive impairments or uniquely contribute to EF deficits ([Baldacara et al., 2024](#); [Butters et al., 2011](#)), excluding studies involving comorbidities limits the generalizability of the findings to real-world clinical populations. Moreover, while studies using symptom-based measures from community samples were only included if they did not report concurrent psychiatric diagnoses, the absence of formal diagnostic assessments means that some degree of undetected comorbidity cannot be entirely ruled out. This introduces some uncertainty regarding the purity of symptom-based samples which may have subtly influenced the observed effects; though, notably, there was no substantial difference in findings between diagnosis-based and symptom/severity-based studies. Nevertheless, further research exploring differences between isolated GAD and GAD with comorbidities will further clarify the unique cognitive profile of GAD, disentangle the specific contributions of GAD versus overlapping conditions to EF impairments, and inform more targeted interventions based on distinct cognitive patterns ([Gkintoni and Ortiz, 2023](#)).

Additionally, the focus on cross-sectional designs in the included studies limits the ability to infer causality between GAD severity and EF deficits. While the meta-analysis confirmed a significant negative association between GAD and EFs, it remains unclear whether EF impairments are a predisposing factor for GAD, a consequence of chronic worry and hypervigilance, or a bidirectional process ([Imran et al., 2017](#); [Mahoney et al., 2018](#)). The stronger EF impairments observed in youth compared to adults suggest that early cognitive vulnerabilities may contribute to the onset of GAD, but the lack of longitudinal studies prevents firm conclusions regarding developmental trajectories ([Kim et al., 2019](#); [Rabner et al., 2024](#)). Many studies also did not account for confounding variables such as sleep quality, medication use, and socioeconomic factors, which are known to affect cognitive functioning and could contribute to EF variability in GAD ([Leonard and Abramovitch, 2019](#); [Trindade et al., 2021](#)). Addressing these gaps will require longitudinal and experimental research to clarify causal mechanisms and evaluate whether cognitive interventions, such as EF training, cognitive-behavioral therapy (CBT), or pharmacological treatments, can mitigate EF impairments in GAD ([Leonard and Abramovitch, 2019](#); [Trindade et al., 2021](#)).

5. Conclusion

This meta-analysis provides evidence that both GAD diagnosis and symptoms/severity is associated with impairments in working memory and cognitive flexibility, but not inhibitory control. Further analyses revealed that GAD significantly impacted working memory (both accuracy and reaction time) and cognitive flexibility (accuracy), suggesting that these domains may be particularly vulnerable to GAD-related dysfunction. This finding challenges attentional control theory, which may not fully account for the broad EF impairments observed in individuals with GAD. Furthermore, GAD appeared to have stronger effects for youths and young/middle-aged adults than older adults, and was consistently linked to poorer self-reported EF across all domains. Further research is needed to clarify the link between GAD and different types of inhibitory control and the developmental trajectory of GAD. Taken together, these findings suggest that GAD has a unique neuropsychological profile that warrants further investigation to better understand which EF domains are most affected and the mechanisms underlying these impairments. Importantly, the results highlight the

potential value of interventions aimed at improving both perceived and objective cognitive abilities. Strategies such as cognitive training targeting working memory and cognitive flexibility, alongside interventions aimed at enhancing confidence in cognitive performance, may offer promising avenues for improving GAD-related cognitive deficits.

CRedit authorship contribution statement

Lan Nguyen: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Jared Walters:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Emily Hutchinson:** Writing – review & editing, Writing – original draft, Investigation. **Yixuan Liu:** Writing – review & editing, Writing – original draft, Investigation. **Xin Li:** Writing – review & editing, Writing – original draft, Investigation. **Chantelle Gudmundsson:** Writing – review & editing, Writing – original draft, Investigation.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2025.119683>.

Data availability

The protocol for this systematic review and meta-analysis was pre-registered in the PROSPERO database (Reference ID: CRD42024591976). Data used within this study are freely available on the Open Science Framework repository: <https://doi.org/10.17605/OSF.IO/7XN6D>.

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